

Lab 1: The Rental Refund Crisis

Excercise 1: Testing the "All or Nothing"

Query

Query History

```
1 SELECT rental_id, customer_id, return_date FROM rental
2 WHERE rental_id = 10;
```

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1

of 1

	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	10	399	2005-05-31 22:44:21

Query

Query History

```
1 BEGIN;
2 UPDATE rental SET return_date = NOW() WHERE rental_id = 10;
3
4 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
5
6 VALUES (459, 1, 10, 99.99, 'WRONG_DATE_FORMAT');
7 COMMIT;
8
9 -- Check the results
10 -- SELECT return_date FROM rental WHERE rental_id=10;
```

Data Output

Messages

Notifications

ROLLBACK

Query returned successfully in 66 msec.

Check the result: We can see that the result isn't changed

Query
Query History

```

1 BEGIN;
2 INSERT INTO customer (first_name, last_name, email, address_id, store_id)
3 VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
4
5 SAVEPOINT user_created;
6
7 INSERT INTO special_offers (customer_id, code) VALUES (9999, 'FREE');
8
9 ROLLBACK TO user_created;
10 COMMIT;

```

Data Output
Messages
Notifications

ROLLBACK

Query returned successfully in 56 msec.

Let's check the result:

Query
Query History

```

1 SELECT first_name, last_name, email, address_id, store_id
2 FROM customer
3 WHERE first_name = 'AI';

```

Data Output
Messages
Notifications

Showing rows: 1 to 1
Page No: 1 of 1

	first_name character varying (45)	last_name character varying (45)	email character varying (50)	address_id smallint	store_id smallint
1	AI	Student	ai@edu.vn	1	1

EXercise 2: The "Flexible" Staff Task

Query
Query History

```

1 SELECT staff_id, first_name, last_name, store_id, username
2 FROM staff
3 WHERE username = 'aistaff';

```

Data Output
Messages
Notifications

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	store_id smallint	username character varying (16)
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Query

Query History

1

BEGIN;

2

3

INSERT INTO staff (first_name, last_name, address_id, email, store_id, username, password)

4

VALUES ('AI', 'Staff', 1, 'aistaff@edu.vn', 1, 'aistaff', '123456');

5

6

SAVEPOINT staff_created;

7

8

UPDATE staff

9

SET store_id = 999

10

WHERE username = 'aistaff';

11

12

ROLLBACK TO staff_created;

13

14

UPDATE staff

15

SET store_id = 1

16

WHERE username = 'aistaff';

17

18

COMMIT;

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